Brian Wonjun Lee

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Education

University of Pennsylvania Philadelphia, PA
MSE in Computer Graphics & Game Technology (CGGT), Submatriculation *Expected May 2028*
BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars *Expected May 2027*
Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Experience

Software Engineer Intern, Aleph Lab (Y Combinator) *June 2026 – Present · San Francisco (Remote)*

- Built a **Python multi-agent tool on the Claude Agent SDK** that mines production data through read-only MCP — used it to show a reported retention gain was a measurement artifact and surface the real churn drivers, **reshaping the team's roadmap**
- Turned the company's production AI agent into a drop-in package other teams can embed in their own apps — a typed SDK with versioned APIs and **136 automated tests**; a new AI coding agent integrated it on the **first attempt**
- Shipped a production retention system (Supabase, Deno edge functions, FCM) that re-engages lapsed users to recover them as paying subscribers, instrumented with a **treatment/holdout A/B**
- Works across the full **LLM product stack** — agent orchestration, MCP tool integration, and config-driven AI modes — at a 4-engineer YC startup

Software Engineer Intern, BitMango (Puzzle1 Studio) *June 2025 – Aug 2025 · Pangyo, South Korea*

- Maintained a live portfolio of **50+ Unity mobile game titles**, improving UI/UX and core gameplay stability across releases
- Resolved **100+ QA-reported bugs** by modifying Unity prefabs and C# scripts, shipping fixes in production releases
- Upgraded SDKs and platform modules fleet-wide to meet evolving app-store requirements and sustain production availability

Project Lead & Product Designer, Penn Spark *Jan 2025 – Present · Philadelphia, PA*

- Led a cross-functional team of **15+** designers and developers to ship client-facing **0-to-1 products**
- Defined end-to-end user flows, interaction patterns, and reusable interface systems across multiple deliverables
- Shipped polished, user-centered features across semester-long client project cycles

Military Interpreter, Capital Defense Command (ROK Army) *Dec 2022 – June 2024 · Seoul, South Korea*

- Provided real-time English–Korean interpretation in high-security, multi-agency environments for international stakeholders
- Translated sensitive documents and briefings with precision, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm *June 2022 – Aug 2022 · Seongnam, South Korea*

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG
- Wrote and optimized SQL queries to improve retrieval and management efficiency across large-scale datasets

Projects

Capsule | *React, Three.js (R3F), MongoDB, AWS S3* *Jan 2025 – Apr 2025*

- Built a **full-stack web app** (team of 4) for creating interactive 3D time capsules with media storage and timed unlocking
- Engineered immersive 3D scenes with React Three Fiber for real-time rendering, animation, and direct-manipulation interaction
- Designed the backend on MongoDB with scalable media storage on AWS S3, wiring authentication and timed-release logic

Mini Minecraft | *C++, OpenGL, GLSL* *Apr 2026*

- Engineered a **voxel engine from scratch**: 7 procedural biomes from layered Perlin/FBM/Voronoi noise with smoothstep blending
- Implemented **3D Perlin cave systems** with trilinear interpolation, fluid physics, and post-process tint overlays
- Built the render pipeline: **7×7 PCF soft shadows, screen-space reflections, vertex AO**, and a dynamic day-night cycle

Mini Maya | *C++, OpenGL, Qt* *Feb 2026 – Mar 2026*

- Implemented a **half-edge mesh data structure** enabling efficient traversal and topology-aware editing operations
- Built edge split, extrusion, and **Catmull–Clark subdivision** on an OpenGL/VBO pipeline with GLSL shaders
- Built an interactive **Qt desktop editor** — vertex/edge/face selection, a GUI control panel, and a real-time OpenGL viewport

Technical Skills

AI & Agents: Claude Agent SDK, MCP, LLM APIs (Anthropic/Gemini), agent orchestration, prompt engineering
Languages: Python, TypeScript, JavaScript, C++, C#, SQL
Frameworks & Tools: React, Next.js, Node.js, Supabase, Deno, Git, GitLab, AWS, MongoDB, Figma